

NETWORKED CONTROL

14/02/2019

Prof. Marcello Farina

EXERCIZES

1) Consider the following continuous-time system $\mathcal{S}$.

$$\begin{cases} \begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 1 \\ \alpha \end{bmatrix} u(t) \\ y(t) = \begin{bmatrix} 1 & \alpha \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} \end{cases}$$

A) Assume that the system $\mathcal{S}$ is controlled using a output-feedback static continuous-time control law of the type $u(t) = -2y(t)$. Analyze the asymptotic stability property of the closed loop system in the following cases:

- a. $\alpha = 1$;
- b. $\alpha = 0$.

B) Compute the parameters F, G , and H of the corresponding discrete-time dynamical equation

$$\begin{aligned} x_{k+1} &= Fx_k + Gu_k \\ y_k &= Hx_k \end{aligned}$$

with sampling time $h = \log_e 2$ s. Also, assuming that the system $\mathcal{S}$ is controlled using a digital control law of the type $u_k = -2y_k$, analyze the asymptotic stability property of the closed loop system in the following cases:

- a. $\alpha = 1$;
- b. $\alpha = 0$.

C) Assume that the system $\mathcal{S}$ is controlled through the networked control system (NCS) sketched in the following figure.

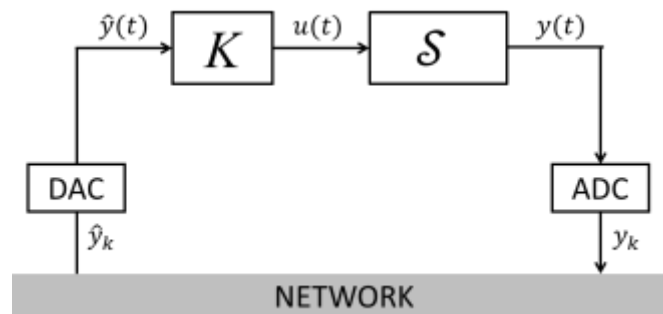


Figure 1: Networked control system

Answer the following questions considering **the case where $\alpha = 0$** .

- a. Define succinctly the MATI (maximal allowable transfer interval) of a NCS.
- b. In case the delay is $\tau = 0$ s, compute the MATI of the NCS.
- c. Is the NCS asymptotically stable in case $h = \log_e 2$ s, the delay is $\tau = 1.5$ s, and in absence of packet dropouts?

SOLUTION

A) Consider the cases $\alpha = 0$ and $\alpha = 1$ separately.

- a. In case $\alpha = 0$, the state variable $x_2(t)$ (that corresponds to mode e^{-2t}) is neither controllable nor observable. For this reason, in this case, to analyze the stability of the closed loop system it is sufficient to analyze the stability properties of the controllable and observable state $x_1(t)$, that evolves according to the equation

$$\dot{x}_1(t) = -x_1(t) + u(t)$$

with $y(t) = x_1(t)$. Setting $u(t) = -2y(t)$, the closed-loop dynamics is

$$\dot{x}_1(t) = -3x_1(t)$$

The closed-loop system is therefore asymptotically stable.

- b. In case $\alpha = 1$, the system is controllable and observable, and no “reduction” can be performed. Setting $u(t) = -2y(t) = -2[1 \ 1] \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$, the closed loop dynamics is described by the equation

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \left(\begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} - 2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} [1 \ 1] \right) \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} -3 & -2 \\ -2 & -4 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

The corresponding characteristic polynomial is $p_{A+BKC}(\lambda) = \lambda^2 + 7\lambda + 8$. The closed-loop system is therefore asymptotically stable.

B) The corresponding discrete-time system is

$$\begin{cases} x_{k+1} &= \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{4} \end{bmatrix} x_k + \begin{bmatrix} \frac{1}{2} \\ \frac{3}{8}\alpha \end{bmatrix} u_k \\ y_k &= [1 \ \alpha] x_k \end{cases}$$

Consider the cases $\alpha = 0$ and $\alpha = 1$ separately.

- a. In case $\alpha = 0$, to analyze the stability of the closed loop system it is sufficient to analyze the stability properties of the controllable and observable state, that evolves according to the equation

$$x_{1,k+1} = \frac{1}{2}x_{1,k} + \frac{1}{2}u_k$$

with $y_k = x_{1,k}$. Setting $u_k = -2x_{1,k}$, the closed-loop dynamics is

$$x_{1,k+1} = -\frac{1}{2}x_{1,k}$$

The closed-loop system is therefore asymptotically stable.

- b. In case $\alpha = 1$, setting $u_k = -2y_k = -2[1 \ 1]x_k$, the closed loop dynamics is described by the equation

$$x_{k+1} = \left(\begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{4} \end{bmatrix} - 2 \begin{bmatrix} \frac{1}{2} \\ \frac{3}{8} \end{bmatrix} [1 \ 1] \right) x_k = \begin{bmatrix} -\frac{1}{2} & -1 \\ -\frac{3}{4} & -\frac{1}{2} \end{bmatrix} x_k$$

The corresponding eigenvalues are $\lambda_1 = 0.366$ and $\lambda_2 = -1.366$. The closed-loop system results unstable.

C) Since now $\alpha = 0$, we restrict our attention to the controllable and observable part of the system, i.e., to equation $\dot{x}_1(t) = -x_1(t) + u(t)$.

- a. **SEE THE LECTURES.**

- b. In case $\tau = 0$, to compute the MATI we can consider the discretized scalar system $x_{1,k+1} = e^{-h}x_{1,k} + (1 - e^{-h})u_k$. Setting $u_k = 2x_{1,k}$, we obtain $x_{1,k+1} = (e^{-h} - 2(1 - e^{-h}))x_{1,k}$. To guarantee asymptotic stability, we need to compute the range of values of h guaranteeing that $|e^{-h} - 2(1 - e^{-h})| < 1$, i.e., $0 < h < \log_e 3$. The MATI is then equal to $\log_e 3$.
- c. In case packet dropouts are absent, but a delay is present, the transition matrix $\Phi(h, \tau)$ is

$$\Phi(h, \tau) = \begin{bmatrix} e^{-h} + \Gamma(h - \tau)K & e^{-(h-\tau)}\Gamma(\tau)K \\ 1 & 0 \end{bmatrix}$$

where $\Gamma(\tau) = \int_0^\tau e^{-\nu} d\nu = (1 - e^{-\tau})$. In case $h = \log_e(2)$ s, $\tau = \log_e 1.5$ s, and $K = -2$,

$$\Phi = \begin{bmatrix} \frac{1}{2} + \frac{1}{4}K & \frac{1}{4}K \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -\frac{1}{2} \\ 1 & 0 \end{bmatrix}$$

whose eigenvalues are $\lambda_{1,2} = \pm j\sqrt{\frac{1}{2}}$, whose modulus is equal to $\sqrt{\frac{1}{2}} < 1$. The NCS is therefore asymptotically stable.

- 2) Consider the general NCS in Figure 1. Assume that the NCS is subject to a time-invariant delay τ and to packet dropouts. Define the asymptotic dropout rate r and state the main theorem that allows to analyze the stability of the NCS with asymptotic dropout rate r **in the deterministic framework**.

SEE THE LECTURES

- 3) Describe the various sources of delays in control networks.

SEE THE LECTURES